PRACTICAL - 1

AIM : Create test cases using boundary value analysis.

Boundary value analysis:

1) Boundary Value Analysis is a software testing method used to test the values at the edges or limits of an input range. It is based on the idea that errors are more likely to happen at boundary values than in the middle of the range.

   Therefore, test cases are created using the minimum value, maximum value, and values just below and above these limits. This technique helps find defects related to input limits and ensures that the system works correctly for edge cases.Date format check

1) Library Book Borrowing Period Check
If books can be borrowed for **1 day - 30 days**.

Invalid	Valid	Invalid
0	1,2,29,30	31

Test Case 1 = 0 (invalid)
Test Case 2 = 1 (valid)
Test Case 3 = 2 (valid)
Test Case 4 = 29 (valid)
Test Case 5 = 30 (valid)
Test Case 6 = 31 (invalid)

Test Scenario ID	**Borrowing Period check**	**Test Case ID**	**Borrowing Period check-1**
Test Case Description	**Verify borrowing period range**	**Test Priority**	**High**
Pre-Requisite	**NA**	**Post-requisite**	**NA**

Test Execution Steps:

S.No	Action	Inputs	Expected Output	Actual Output	Test Result
1	Borrowing Period check	2 Days	Valid	Valid	Pass
2	Borrowing Period check	31 Days	Invalid	Invalid	Fail

2) Hotel Floor Selection Check

If a hotel has floors ranging from **1 - 20**.

Invalid	**Valid**	**Invalid**
0	1,2,19,20	21

Test Case 1 = 0 (invalid)

Test Case 2 = 1 (valid)

Test Case 3 = 2 (valid)

Test Case 4 = 19 (valid)

Test Case 5 = 20 (valid)

Test Case 6 = 21 (invalid)

Test Scenario ID	**Floor check**	**Test Case ID**	**Floor check-1**
Test Case Description	Verify floor number range	Test Priority	High
Pre-Requisite	NA	Post-requisite	NA

Test

Execution Steps:

S.No	**Action**	**Inputs**	**Expected Output**	**Actual Output**	**Test Result**
1	Floor check	Floor 2	Valid	Valid	Pass
2	Floor check	Floor 21	Invalid	Invalid	Fail

3)Online Meeting Duration Check
If an online meeting duration should be between **15 - 180 minutes**.

Invalid	**Valid**	**Invalid**
14	15,16,179,180	181

Test Case 1 = 14 (invalid)

Test Case 2 = 15 (valid)

Test Case 3 = 16 (valid)

Test Case 4 = 179 (valid)

Test Case 5 = 180 (valid)

Test Case 6 = 181 (invalid)

Test Scenario ID	**Meeting Duration check**	**Test Case ID**	**Meeting Duration check-1**
Test Case Description	Verify meeting duration range	Test Priority	High
Pre-Requisite	NA	Post-requisite	NA

Test Execution Steps:

S.No	Action	Inputs	Expected Output	Actual Output	Test Result
1	Meeting Duration check	16 Minutes	Valid	Valid	Pass
2	Meeting Duration check	181 Minutes	Invalid	Invalid	Fail

4) Cinema Seat Number Check
If seat numbers are available from **1 - 200**.

Invalid	**Valid**	**Invalid**
0	1,2,199,200	201

Test Case 1 = 0 (invalid)
Test Case 2 = 1 (valid)
Test Case 3 = 2 (valid)
Test Case 4 = 199 (valid)
Test Case 5 = 200 (valid)
Test Case 6 = 201 (invalid)

Test Scenario ID	**Seat Number check**	**Test Case ID**	**Seat Number check-1**
Test Case Description	Verify seat number range	Test Priority	High
Pre-Requisite	NA	Post-requisite	NA

Test Execution Steps:

S.No	Action	Inputs	Expected Output	Actual Output	Test Result
1	Seat Number check	Seat No. 2	Valid	Valid	Pass
2	Seat Number check	Seat No. 201	Invalid	Invalid	Fail